

Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the air supply to the air starter to prevent accidental engine starting, if installed.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the crankcase breathers. Refer to Procedure 003-001 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html)
- Remove the fuel supply manifold. Refer to Procedure 006-022 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-022-tr.html)
- Remove the fuel drain lines. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-013-tr.html)
- Remove the exhaust manifold capscrews from the cylinder head being removed. Loosen the exhaust manifold capscrews from the other heads on the same manifold section. Refer to

Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html>)

- Disconnect the wiring harness in the areas that restrict access to the cylinder head. Use this procedure from the Troubleshooting and Repair Manual, QSK38, QSK50 and QSK60 , Bulletin 4021533. Refer to Procedure 019-043 in Section 19.
(</qs3/pubsys2/xml/en/procedures/05/05-019-043.html>)
- Remove the air crossover. Refer to Procedure 010-002 in Section 10.
(</qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html>)
- Remove the aftercooler assembly. Refer to Procedure 010-002 in Section 10.
(</qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html>)
- Remove the fuel manifold. Refer to Procedure 006-022 in Section 6.
(</qs3/pubsys2/xml/en/procedures/56/56-006-022-tr.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(</qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html>)
- Remove the rocker levers. Refer to Procedure 003-009 in Section 3.
(</qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html>)
- Remove the crossheads. Refer to Procedure 002-001 in Section 2.
(</qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html>)
- Remove the push rods. Refer to Procedure 004-014 in Section 4.
(</qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html>)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3.
(</qs3/pubsys2/xml/en/procedures/56/56-003-013-tr.html>)
- Remove the injectors. Refer to Procedure 006-026 in Section 6.
(</qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html>)

QSK60 Marine Applications

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Note : On two-loop cooling systems, drain both of the cooling systems.

- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the air supply to the air starter to prevent accidental engine starting, if installed.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the crankcase breathers. Refer to Procedure 003-001 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html)
- Remove the aftercooler heat shields. Refer to Procedure 010-129 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html)
- Remove the turbocharger heat shields. Refer to Procedure 010-076 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html)
- Remove the exhaust manifold heat shields. Refer to Procedure 011-015 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-015.html)
- Remove the exhaust manifold capscrews from the cylinder head being removed. Loosen the exhaust manifold capscrews from the other heads on the same manifold section. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html)
- Disconnect the wiring harness in the areas that restrict access to the cylinder head. Use this procedure from the Troubleshooting and Repair Manual, QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Bulletin 4021533. Refer to Procedure 019-043 in Section 19.
(/qs3/pubsys2/xml/en/procedures/05/05-019-043.html)
- Remove the air crossover. Refer to Procedure 010-002 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html)
- Remove the aftercooler assembly. Refer to Procedure 010-002 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html)
- Remove the fuel manifold. Refer to Procedure 006-022 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-022-tr.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html)
- Remove the rocker levers. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html)
- Remove the crossheads. Refer to Procedure 002-001 in Section 2.
(/qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html)
- Remove the push rods. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-013-tr.html)
- Remove the injectors. Refer to Procedure 006-026 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)

with Electronically Actuated Injector

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

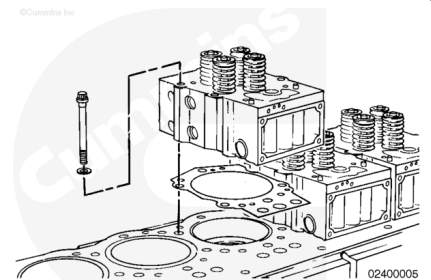
- Disconnect the batteries. See equipment manufacturer service information.
- Disconnect the air supply to the air starter to prevent accidental engine starting, if installed.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the crankcase breathers. Refer to Procedure 003-001 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-001-tr.html)
- Remove the injector supply lines. Refer to Procedure 006-051 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-051-tr.html)
- Remove the exhaust manifold capscrews from the cylinder head being removed. Loosen the exhaust manifold capscrews from the other heads on the same manifold section. Refer to Procedure 011-007 in Section 11. (/qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html)
- Disconnect the wiring harness in the areas that restrict access to the cylinder head. Use this procedure from the Troubleshooting and Repair Manual, QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Bulletin 4021533. Refer to Procedure 019-043 in Section 19.
(/qs3/pubsys2/xml/en/procedures/05/05-019-043.html)
- Remove the air crossover. Refer to Procedure 010-002 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html)
- Remove the aftercooler assembly. Refer to Procedure 010-002 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html)

- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html)
- Remove the injector. Refer to Procedure 006-026 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)
- Remove the rocker levers. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html)
- Remove the crossheads. Refer to Procedure 002-001 in Section 2.
(/qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html)
- Remove the push rods. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-013-tr.html)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the seven cylinder head mounting capscrews.

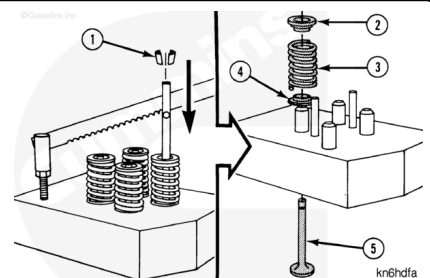
Remove the cylinder head.

Remove and discard the cylinder head gasket.

Disassemble

⚠ WARNING ⚠

The valve springs are under compression. Use caution when using the valve spring compressor to reduce the possibility of personal injury.



Note : Cummins Inc. recommends using new valve springs when the cylinder head is reconditioned.

Use a spring compressor, Part Number 3163606, or Part Number 3375960, or equivalent. A valve spring stand, Part Number ST-1022, and the compressor plate, Part Number 3375043, can be used to compress all four springs at the same time.

Mark the valves for location prior to removal. The intake and exhaust valves are different. This practice will aid the assembly procedure.

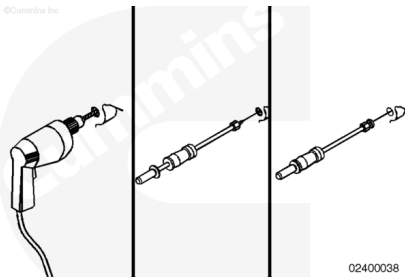
Remove the following parts:

- 1. Valve collet
- 2. Valve spring retainer
- 3. Valve spring
- 4. Valve rotator
- 5. Valve.

Discard the valve springs, collets, and valve rotators.

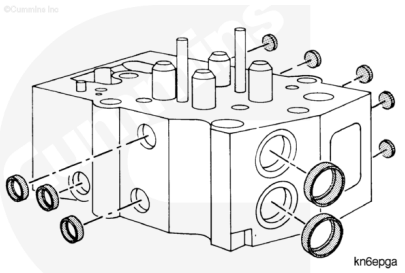
Use a drill, sheet metal screw, and the following parts from the light-duty puller kit, Part Number 3375784, to remove the cup plugs.

- Slide hammer
- Hook.



Remove and discard the ten cup plugs.

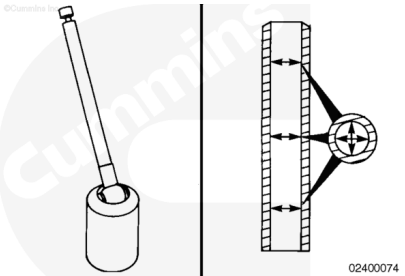
The cup plugs **must** be removed from the head casting for cleaning purposes.



Use a ball gauge or a dial bore indicator to measure the inside diameter of the valve guides.

Measure the inside diameter.

Valve Guide Inside Diameter (Installed)		
mm		in
10.561	MIN	0.4158
10.589	MAX	0.4169

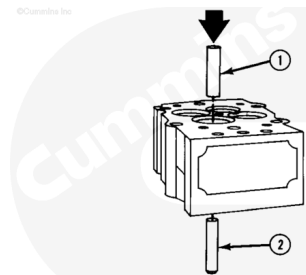


⚠ CAUTION ⚠

If a valve guide is replaced, the valve seat insert in that location must be ground or severe engine damage can result.

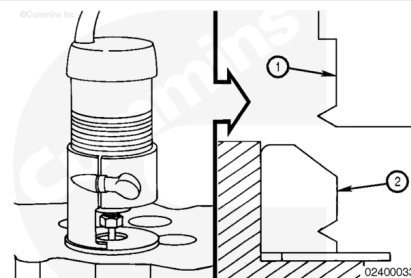
Only replace the valve guide if it is **not** within specifications.

Use an arbor press and a suitable mandrel (1) to remove the worn valve guide (2).



Use the valve seat grooving kit, Part Number 3376405, or equivalent.

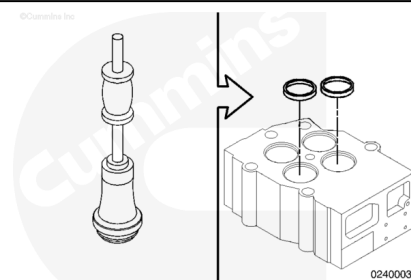
Machine a groove in the seat as close to the bottom of the bore as possible. This will allow the valve seat extractor to be used.



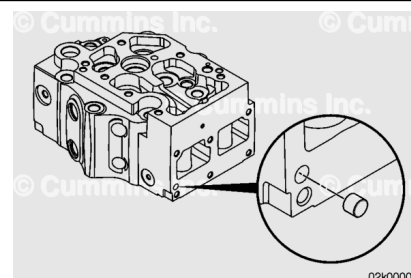
Use the specified intake and exhaust valve seat extractor. Use an extractor knocker assembly, Part Number 3376617, to remove the valve seats.

- Exhaust seat extractor, Part Number ST-323-1
- Intake seat extractor, Part Number 3376799.

Discard the anti-swirl plates that are contained **only** in the intake ports.



Remove the existing tamper proof seal from the intake face of the cylinder head. Use service tool, Part Number 3823154, or equivalent.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the cylinder heads, capscrews, valve spring retainers, and valves with solvent.

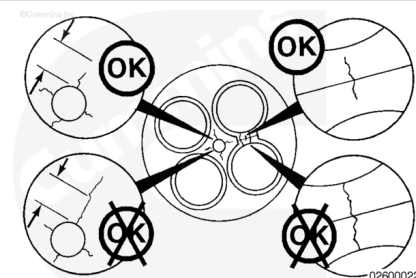
Note : To prevent damage to the capscrew coating, do not use a wire brush or any abrasive cleaning method on the cylinder head capscrews.

Dry with compressed air.

Check for cracks on the combustion surface.

If a crack around the injector bore exceeds the maximum length, the cylinder head **must** be replaced.

Injector Bore Allowable Crack Length		
mm		in
6.4	MAX	0.252

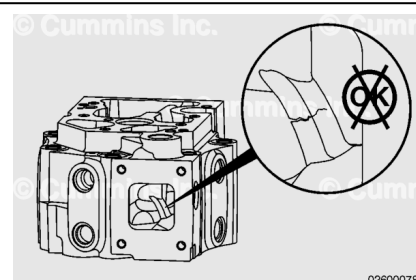


Both ends of a crack between the valve **must** be visible. If one crack extends into the valve seat bore (behind the valve seat), the condition of the cylinder head is questionable.

Check for cracks in the exhaust port divider perpendicular to the combustion face toward the long port exhaust valve.

If a crack in the exhaust port divider exceeds the maximum length, the cylinder head **must** be replaced.

Exhaust Port Divider Allowable Crack Length		
mm		in
15	MAX	0.59

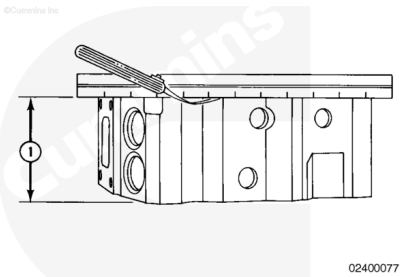


Use a straight edge, Part Number 4918219, or equivalent, to measure the flatness of the head gasket surface.

The head **must** be resurfaced if a feeler gauge larger than 0.08 mm [0.003 in] will fit between the straight edge and the cylinder head.

Note : Cylinder heads with electronically actuated injectors can not be resurfaced since correct injector protrusion can not be maintained after resurfacing.

The head can be resurfaced if head thickness (1) measurement is within specifications.



New Cylinder Head

mm		in
150.876	MIN	5.940
151.130	MAX	5.950

Remanufactured Cylinder Head

mm		in
150.114	MIN	5.910
151.130	MAX	5.950

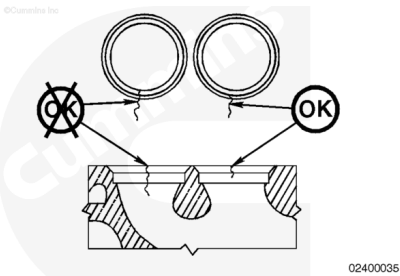
If the head is resurfaced, make certain the injector protrusion and valve depth in the head are adjusted properly.

Clean the valve seat insert bore.

Check the length of any cracks extending into the valve seat insert bore.

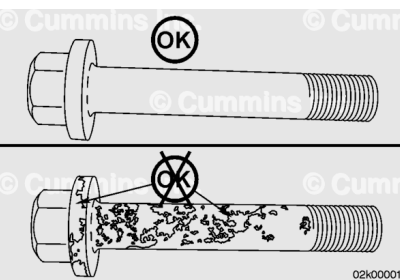
The cylinder head **must** be replaced if a crack extends into the bottom of the bore.

Sometimes it is possible to remove the crack by machining the head to use an oversize valve seat insert.



Inspect the cylinder head capscrew coating, especially under the capscrew head and at the threads, for nicks, flakes, or missing spots.

If the capscrew coating is damaged or missing, the capscrew **must** be replaced.

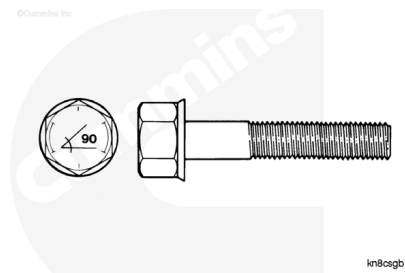


The use of the torque plus angle method of installation places the cylinder head capscrew beyond the yield point and permanently stretches the capscrew. These capscrews can be

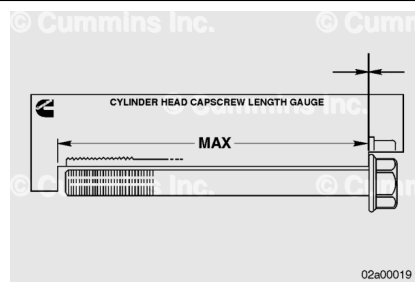
reused throughout the life of the engine unless the capscrew exceeds the specified free length.

The free length **must** be checked to avoid bottoming in the block during installation and to make sure the capscrew still has the proper strength.

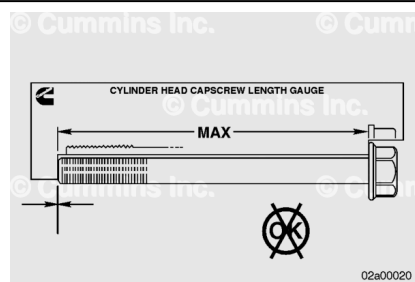
A cylinder head capscrew length gauge, Part Number 3163328, has been developed to check the capscrew free length.



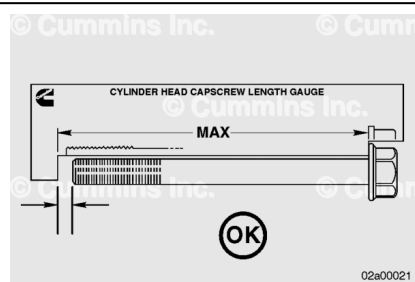
To check the capscrew free length, place the head of the capscrew in the slot with the flange against the base of the slot.



If the end of the capscrew touches the foot of the gauge, the capscrew is too long and **must** be discarded.

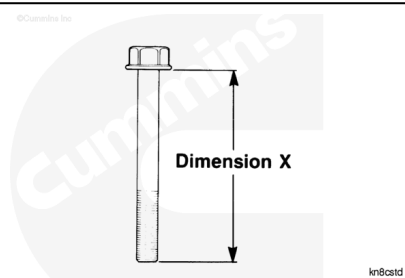


If there is clearance between the end of the capscrew and the bottom base of the tool, the capscrew is acceptable for reuse.

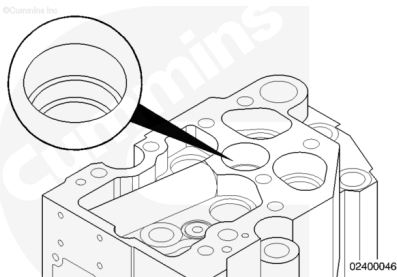


The capscrew can also be checked using a set of calipers. The maximum allowable free length is measured from the bottom of the flange to the end of the capscrew (dimension X).

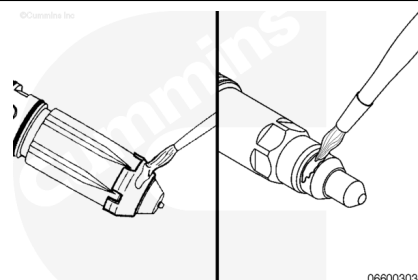
Cylinder Head Capscrew Free Length		
mm		in
202.5	MAX	7.972



Inspect the injector bore for cracks or other damage.

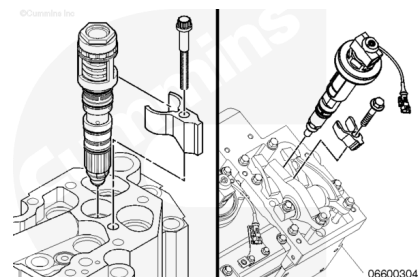


Apply a thin coating of Prussian blue on the sealing ring.



⚠ CAUTION ⚠

Support the cylinder head to reduce the risk of damage to the injector tip that protrudes from the combustion face.



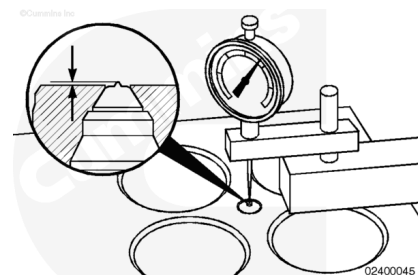
Install the injector and injector hold-down clamp.

Tighten the hold-down capscrew.

Torque Value: 16 n•m [142 in-lb]

Use a gauge block, Part Number 3164438, or equivalent, to measure the injector protrusion.

Measure as shown in the illustration.



Electronically Actuated Injector (MCRS) - Protrusion

mm		in
2.82	MIN	0.111
3.22	MAX	0.127

Note : Injector protrusion can not be changed on engines with electronically actuated injectors.

Mechanically Actuated Injector (HPI) - Protrusion

mm		in
2.2	MIN	0.087
2.7	MAX	0.106

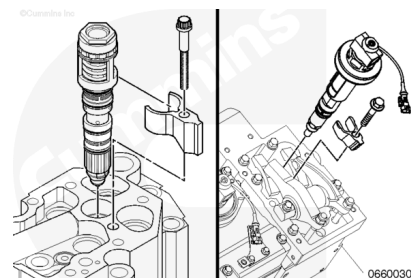
If the protrusion is **not** within specifications, use an oversize sealing ring. Sealing rings are available in various sizes.

Injector Sealing Rings			
	mm		in
Wall thickness	0.343	MIN	0.0135
	0.419	MAX	0.0165
Protrusion change	0.00	NOM	0.000
Wall thickness	0.470	MIN	0.0185
	0.546	MAX	0.0215
Protrusion change	0.25	NOM	0.010
Wall thickness	0.597	MIN	0.0235
	0.673	MAX	0.0265
Protrusion change	0.51	NOM	0.020
Wall thickness	0.724	MIN	0.0285
	0.800	MAX	0.0315
Protrusion change	0.76	NOM	0.030

The injector sealing ring with a wall thickness of 0.343 to 0.419 mm [0.0135 to 0.0165 in] is the standard sealing ring.

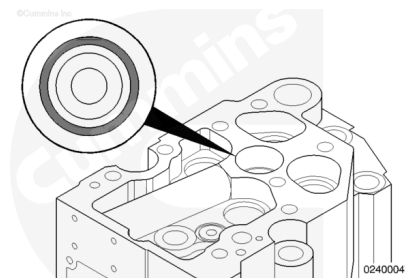
Remove the injector hold-down clamp capscrew.

Remove the injector and injector hold-down clamp.



Inspect the contact pattern in the injector bore. A blue band 1.52 mm [0.060 in] wide and 360 degrees full circumference minimum **must** be visible.

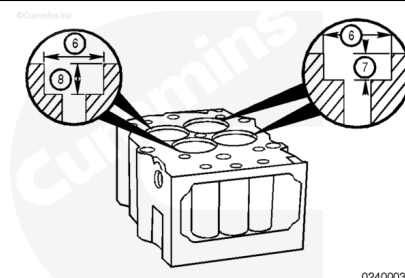
If the pattern does **not** meet the specification, clean the injector bore with a tapered brush and inspect the seating pattern again. The cylinder head **must** be replaced if it is **not** within specifications.



With the valve seats removed, measure the inside diameter and the depth of the insert bore in the head.

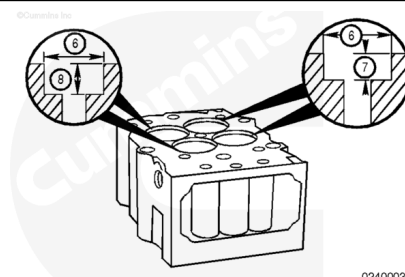
Valve Seat Bore Measurements

	mm		in
Intake diameter	60.37	MIN	2.377
	60.40	MAX	2.378
Intake depth	13.00	MIN	0.512
	13.13	MAX	0.517
Exhaust diameter	60.37	MIN	2.377
	60.40	MAX	3.378
Exhaust depth	12.50	MIN	0.492
	12.62	MAX	0.497



If the inside diameter is **not** within specifications, use an oversize seat.

Oversize valve seat inserts are available in the following sizes. Machine the valve seat insert bore in the head. A press fit within 0.064 to 0.114 mm [0.0025 to 0.0045 in] between the valve seat insert and the valve seat insert bore **must** be maintained.



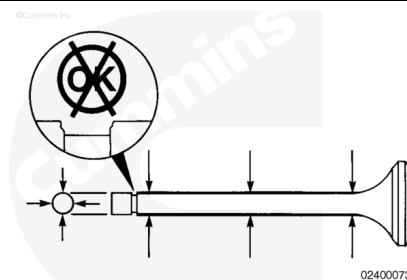
Outside Diameter Oversize	Depth (Thickness) Oversize
0.25 mm [0.010 in]	Standard
0.51 mm [0.020 in]	0.13 mm [0.005 in]
0.76 mm [0.030 in]	0.25 mm [0.010 in]
1.02 mm [0.040 in]	0.38 mm [0.015 in]

Note : For the QSK60 G3 and G3 engines, oversize seats are **not** currently available.

Inspect the collet grooves for wear.

Measure the outside diameter of the valve stem.

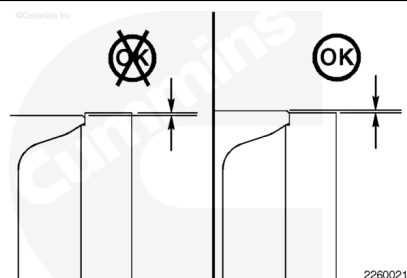
Valve Stem Outside Diameter		
mm		in
10.490	MIN	0.4130
10.510	MAX	0.4138



Use the valve head checking tool, Part Number 3164984, to measure the head thickness of the valve.

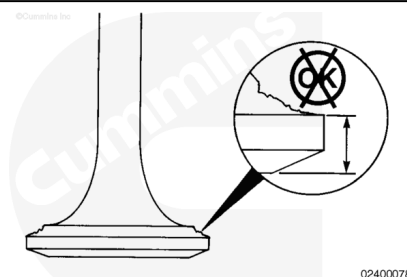
Place the valve into the valve head thickness gauge.

Measure the valve height. If the valve is flush or above the gauge surface, the valve can be reused. If the valve is below the gauging surface, the valve can **not** be reused.



If a valve head checking tool is **not** available, put the valve on a flat surface. Check the height of the outside diameter.

Valve Head Thickness (At the Outside Diameter)		
mm		in
3.94	MIN	0.155

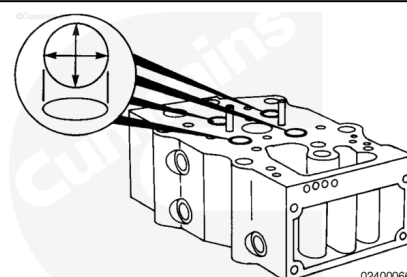


This type of a check is **not** as accurate as using the valve head checking tool.

Measure the inside diameter of the valve guide bore.

If the bore is **not** within specifications, ream the bore for oversize valve guides.

Valve Guide Bore Inside Diameter		
mm		in
21.438	MIN	0.8440
21.463	MAX	0.8450



Valve guides are available in two oversizes: 0.25 mm [0.010 in] and 0.38 mm [0.015 in]. Ream the valve guide bore to maintain a press fit with 0.28 and 0.067 mm [0.00111 and 0.0026 in] between the valve guide and the valve guide bore.

Note : Cummins Inc. recommends using new valve springs when the cylinder head is reconditioned.

A bent or broken spring **must not** be used again.

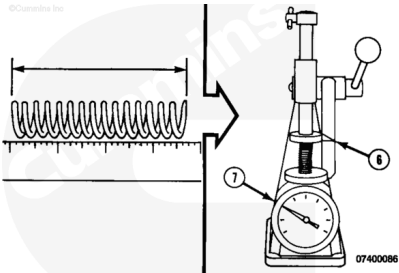
A spring with a notch worn in the first coil **must not** be used again. The ends of the springs will wear into the first coil, creating a worn notch in the material.

If a worn notch can be seen or felt, the valve spring **must not** be used again.

Often a spring that has these notches will make a clicking sound. Hand compress the spring until the end of the first coil is completely collapsed to check for a click sound. A large notch will cause a clicking sound.

Check both ends of the spring for the clicking sound.

Check the spring free length.



Measurements

	mm	in
Valve Spring Approximate Free Length	87.63	3.45

Use valve spring compressor tool, Part Number 3375182, to check the spring force at the indicated test height.

Test Height and Valve Spring Force			
	n		lbf
71.88 mm [2.83 in]	716	MIN	161
	796	MAX	179
57.23 mm [2.253 in]	1477	MIN	332
	1637	MAX	368

If the valve spring is **not** within specification, the valve spring **must** be replaced.

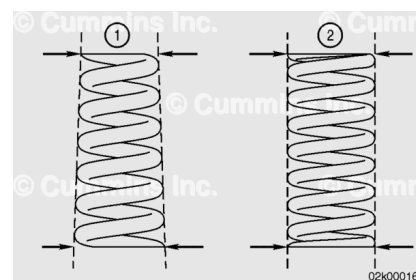
Check the spring free length.

Valve Spring Approximate Free Length

Measurements - Old Style Spring (1)

	mm	in
Valve Spring Approximate Free Length	87.63	3.45

New Style Spring (2)		
	mm	in
Valve Spring Approximate Free Length	91.9	3.62



Use valve spring compressor tool, Part Number 3375182, to check the spring force at the indicated test height.

Test Height and Valve Spring Force - Old Style Spring (1)

Test Height	n		lbf
71.88 mm [2.830 in]	716	Min	161
	796	Max	179
57.23 mm [2.253 in]	1477	Min	332
	1637	Max	368

Test Height and Valve Spring Force - New Style Spring (2)

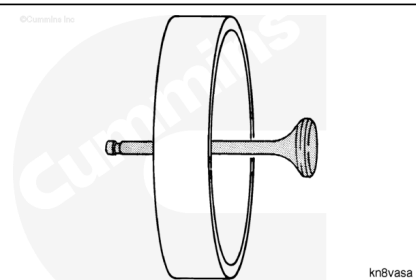
Test Height	n		lbf
71.88 mm [2.830 in]	800	Min	180
	890	Max	200
57.23 mm [2.253 in]	1477	Min	332
	1637	Max	368

If the valve spring is **not** within specification, the valve spring **must** be replaced.

Magnetic Crack Inspect

⚠ CAUTION ⚠

Always demagnetize and clean the parts thoroughly after a magnetic particle inspection. The iron fragments can damage the internal engine components.



Use the magnetic particle residual method.

Check the valves for cracks.

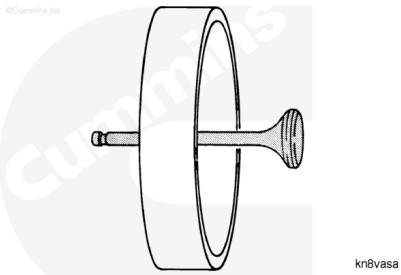
Check the intake valves with the coil and head shot methods (the coil shot first). Check the exhaust valves with the coil shot method.

Use a 305 mm [12 in] minimum diameter coil.

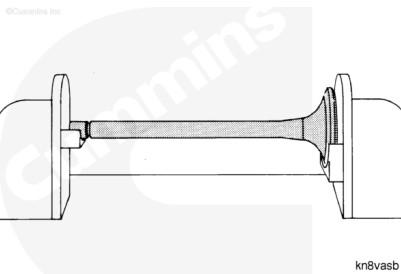
Coil Shot Amperage (Ampere Turns)	
Minimum	Maximum
400-VDC	800-VDC

An ampere turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

A broad, fuzzy pattern will appear at the welded joint on the exhaust valves. This is normal. If there is a distinct line in the broad, fuzzy pattern, the valve **must** be replaced.

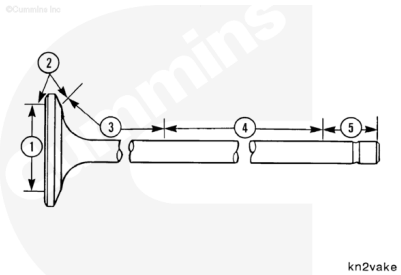


Head Shot Amperage (Ampere)	
Minimum	Maximum
500 VDC	700 VDC



Acceptance criteria (intake and exhaust valves):

1. Indications less than 38.1 mm [1.5 in] in length are acceptable. More than five indications, spaced closer than 3 mm [0.118 in], are **not** acceptable.
2. No indications are acceptable.
3. **Only** longitudinal indications are acceptable.
4. No indications are acceptable.



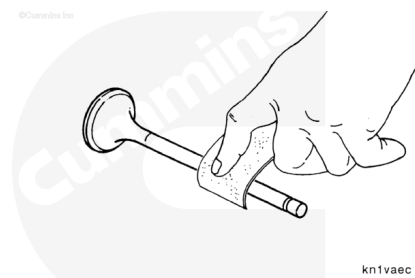
Grind

Valves

Do **not** use aluminum oxide cloth or sandpaper to clean the valves.

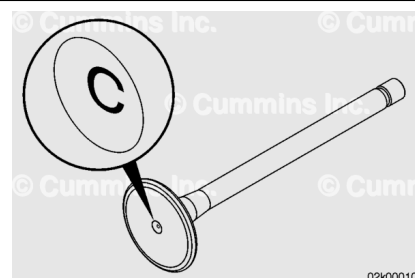
Use Scotch-Brite™, Part Number 3823258, to clean the valve stems. Clean the carbon deposits from the valve face and the head.

The valves **must** be clean and free of carbon deposits before they are ground. Valves can be cleaned by the bead blasting method in the head area **only**. If the stems are bead blasted, the valve **must not** be reused, because the surface finish will be damaged.



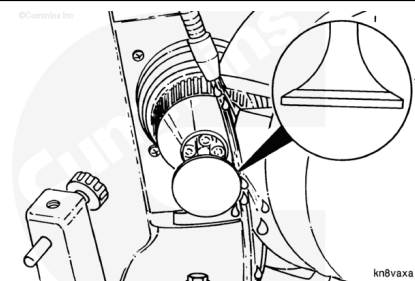
⚠ CAUTION ⚠

Valves marked with an "A, C or U" in the center of the valve head, **must not be ground**. Grinding these valves will damage the hardened layer of the valve face and severe engine damage can result.



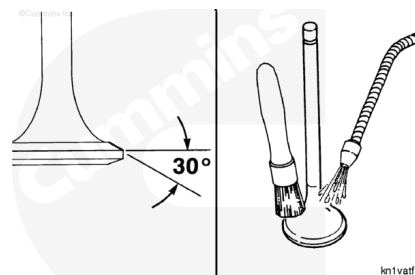
Use a valve facing machine, Part Number 3376256, or equivalent, to grind the face of the valve.

Follow the instructions supplied with the valve facing machine for the correct setup before grinding the valves.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

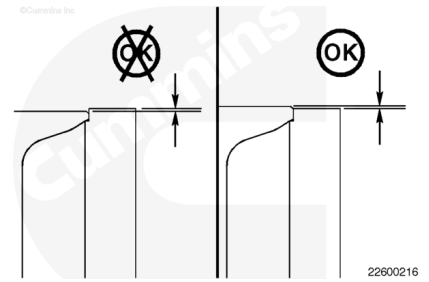
Use solvent to clean the metal particles from the valve.

Dry with compressed air.

Use the valve head checking tool, Part Number 3164984, to measure the head thickness of the valve.

Place the valve into the valve head thickness gauge.

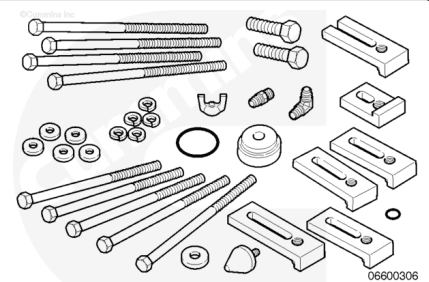
Measure the valve height. If the valve is flush or above the gauge surface, the valve can be reused. If the valve is below the gauging surface, the valve can **not** be reused.



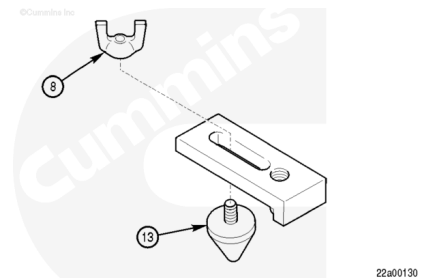
Pressure Test

Preferred Method

Use the Cylinder Head Leak Test Kit, Part Number 3164341, to pressure test the cylinder head.



Assemble the wing nut (8), clamp, and plug (13), as necessary.

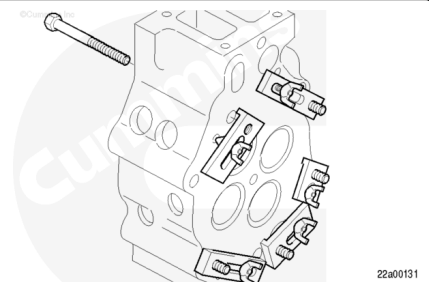


Install the clamp and plug assembly into the cylinder head, blocking the coolant passages.

Insert the appropriate capscrew through the valve guide side of the cylinder head, and thread it into the clamp.

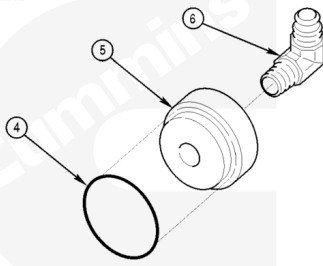
Tighten the capscrews.

Torque Value: 20 n·m [177 in-lb]



Install the o-ring (4) into the o-ring groove of the test adapter (5).

Thread the adapter elbow (6) into the adapter (5).

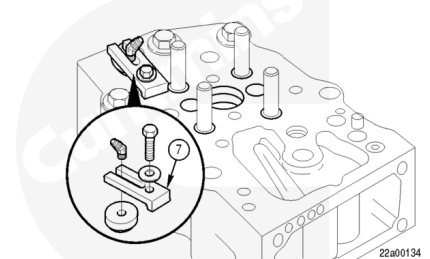


22A00133

Place the test adapter assembly into the cylinder head.

Install the clamp (7) over the test adapter assembly and secure it with the appropriate capscrew and washer.

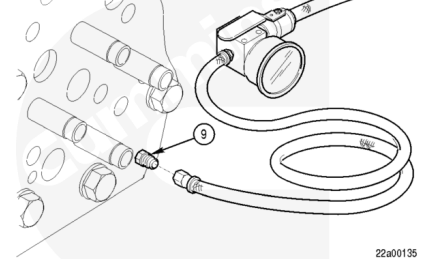
Torque Value: 54 N·m [40 ft-lb]



22A00134

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



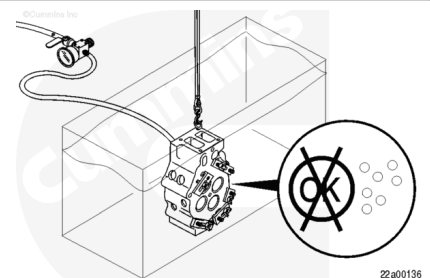
22A00135

Connect the pressure regulator valve kit to the pipe nipple (9).

Connect the pressure regulator valve kit to shop air and adjust the air pressure to 345 kPa [50 psi].

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



22A00136

Note : The QST30 series engine cylinder head is shown.

Attach an eyebolt to the cylinder head.

Use a suitable lifting device to lift the cylinder head.

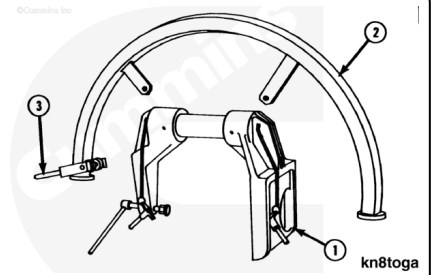
Immerse the cylinder head into a tank of water. Check for bubbles. Bubbles indicate an air leak.

If the cylinder head leaks, it **must** be repaired or replaced.

Optional Method

⚠ CAUTION ⚠

Hydrostatic tester, Part Number ST-1012, must not be used on cylinder heads with electronically actuated injectors. The larger injector bore will be damaged if this tool is used.

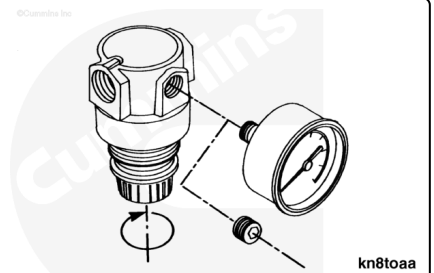


Use the hydrostatic tester, Part Number ST-1012, with water test adapter plate, Part Number 3375070, to test the cylinder head.

Remove one of the two plugs from the pressure regulator.

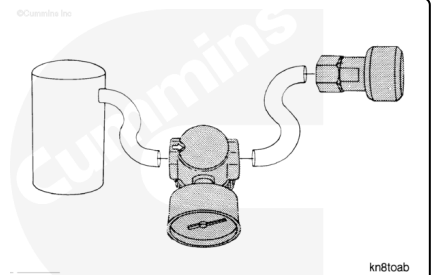
Install the pressure gauge in the regulator.

Turn the adjusting knob on the regulator, **counterclockwise**, as far as it will turn.

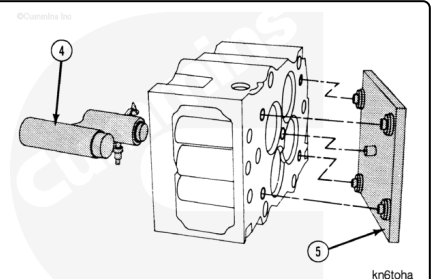


Install the pressure regulator between the air supply and the quick disconnect fitting.

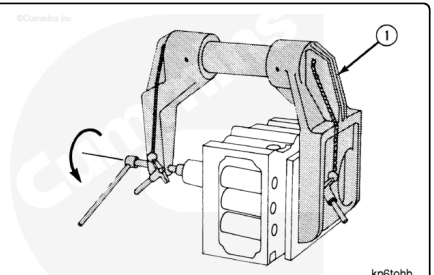
The arrow on top of the pressure regulator **must** point in the direction of the airflow (toward the quick disconnect fitting).



Use adapter (4), Part Number 3375070, and plate (5), Part Number 3375071, to test the head.



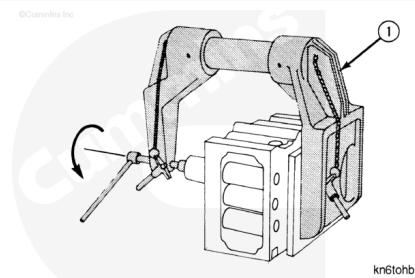
With the head as shown, assemble the adapter plates. The guide pins on the lower plate **must** go in the water passages. The o-ring on the upper plate creates a seal on the upper water passage. The other end of the plate goes into the injector bore.



Place the clamping assembly (1) over the head and adapters. The guide pins on the clamp **must** go into the holes in the lower adapter plate.

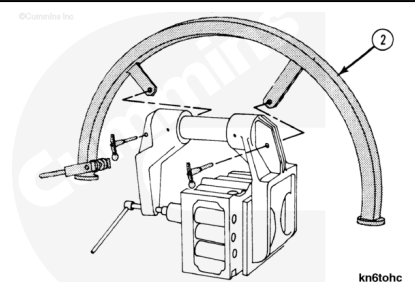
Tighten the clamp on the head.

Connect the wire hose to the upper adapter fitting.



Attach the lifting arm (2) to the clamp assembly with the lock pins.

One mounting location on each piece is color coded red.

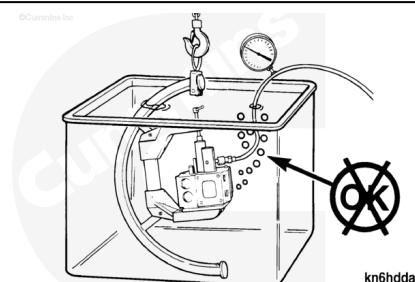


Adjust the air pressure to 552 kPa [80 psi].

Lower the head into a tank of water.

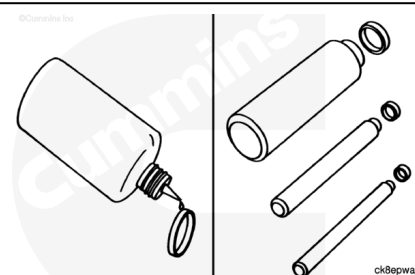
Bubbles will indicate an air leak.

If the cylinder head leaks, it **must** be repaired or replaced.



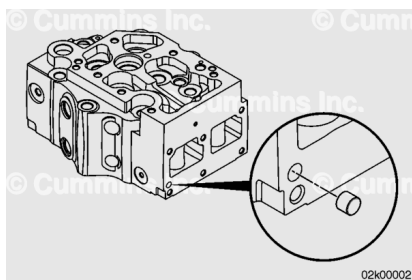
Assemble

Use cup plug Loctite™ sealant, Part Number 3375068, for cup plug installation.



Install a tamper proof seal into the intake face of the cylinder head. Use service tool, Part Number 3823154, or equivalent.

Note : The tamper proof seal must not protrude out of the casting.

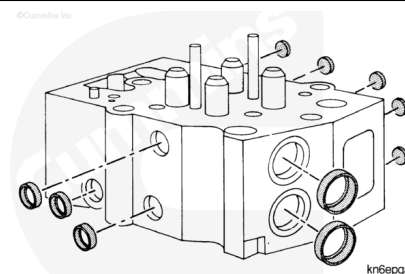


Use the correct expansion plug driver to make sure the expansion plugs are installed to the correct depth in the cylinder head. Drive the plug until the shoulder of the driver contacts the cylinder head.

- Expansion plug driver, Part Number 3376813
- Expansion plug driver, Part Number 3376814.

The cup plug driver handle, Part Number 3164085 or 3376795, **must** be used with the appropriate cup plug driver head, Part Number 3376813 or 3376814.

Install 10 new cup plugs.

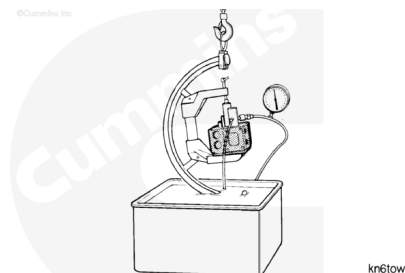


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a hydrostatic tester to check for evidence of leaks around the cup plugs.

Dry the cylinder head with compressed air after testing.



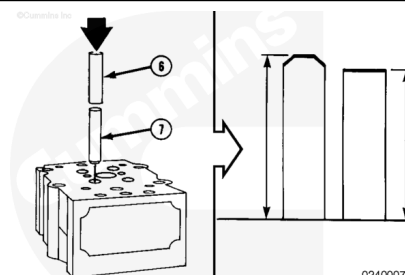
⚠ CAUTION ⚠

Make certain the mandrel does not damage the top of the valve guide. If the guide is damaged, the valves can stick, causing severe engine damage.

Install the valve guides in all four locations in the cylinder head.

Use valve guide driver (6), Part Number 3163041, and an arbor press to install the valve guide (7) to the specified height.

Install the valve guide (7) to the specified height.

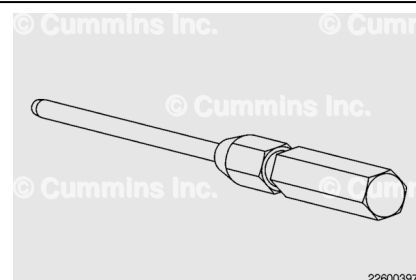


Valve Guide Height			
	mm		in
To spring pocket	37.97	MIN	1.495
	38.48	MAX	1.515
To top of head	13.97	MIN	0.550
	14.47	MAX	0.570

Inspect the valve guides for chips and cracks.

Use a drop gauge, Part Number 5298508, to check the valve guide for proper inside diameter and straightness.

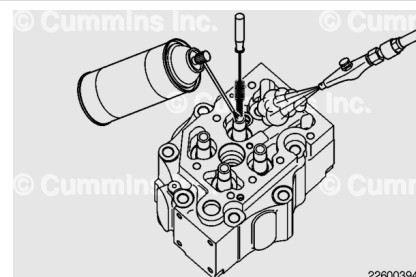
Note : The drop gauge must be adjusted to a minimum of 114 mm [4.5 in] long.



⚠ WARNING ⚠

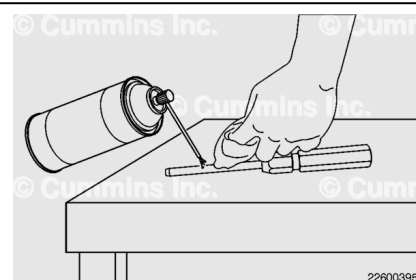
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

- Clean the valve guide bores with an aerosol solvent (such as brake cleaner or carburetor cleaner) and a 7/16 inch diameter nylon brush.
- Use compressed air to dry the valve guide bores.
- Orient the cylinder head with the combustion face down, sitting flat on a level surface.



Clean the drop gauge pin. Use the same type of solvent that is used on the valve guide bores.

Note : Inspect the drop gauge pin for nicks, dings, scratches, and abrasion that can cause a false failure on the drop gauge test. Excessive wear can cause a false pass.

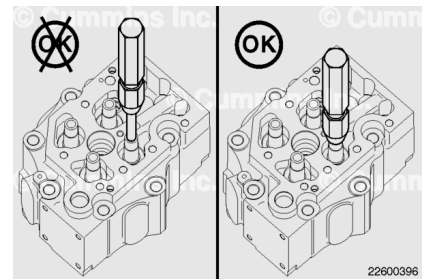


Holding the drop gauge by the upper end of the handle, place the tip of the gauge pin in the bore and let the gauge drop freely, by gravity alone.

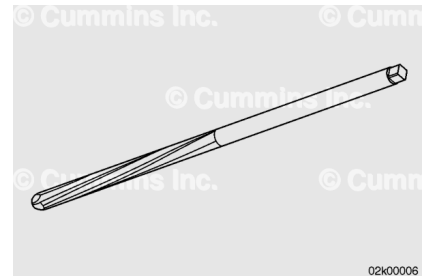
The valve guide passes if the gauge falls until the base of the handle contacts the valve guide.

If the drop gauge does **not** fall until the base of the tool contacts the valve guide, rotate the tool 90 degrees and test again.

If the drop gauge test fails at all four positions, the guide **must** be reamed and tested again.

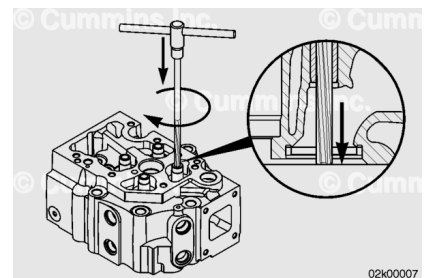


When a valve guide does **not** pass the drop gauge test, it **must** be reamed to bring the bore to the correct specification. The reamer, Part Number 5298510, is the **only** reamer to be used for this purpose. The reamer is to be used **only** by hand with a hand 'T' bar. It is **not** to be driven by any type of power drill or milling machine.

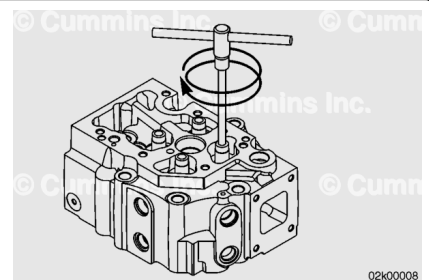


Note : It is essential that the reamer is continuously rotated while it is being inserted or removed from the valve guide. Damage to the valve guide will occur if the reamer is inserted or removed without rotation.

Position the cylinder head with the combustion face down. Insert the tip of the reamer into the top of the valve guide. While rotating the reamer **clockwise**, allow it to drop by gravity into the valve guide bore. Continue to rotate the reamer by hand, in a **clockwise** direction, until it drops through the valve guide and rests on the surface below.

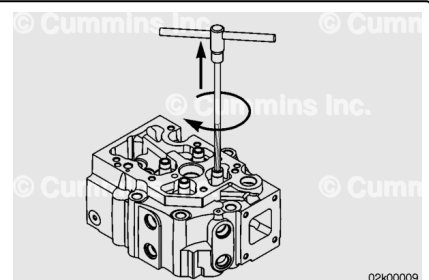


Once the reamer has dropped through the valve guide, rotate it through approximately two complete turns, continuing to rotate the reamer by hand, in a **clockwise** direction.

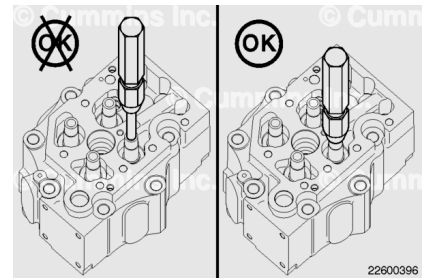


Note : It is essential that the reamer is continuously rotated while it is being inserted or removed from the valve guide. Damage to the valve guide will occur if the reamer is inserted or removed without rotation.

Withdraw the reamer while continuing to rotate it in a **clockwise** direction.

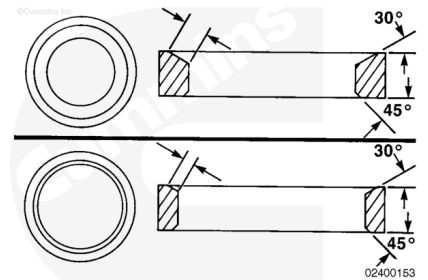


Make sure the valve guide is thoroughly cleaned. Repeat the drop gauge test as described above. If a valve guide does **not** pass the drop gauge test, repeat the reamer process until a drop gauge pass is achieved.



Identify the correct valve seat insert.

- The top illustration is the intake port valve seat insert with a wide valve seating area.
- The bottom illustration is the exhaust port valve seat insert.

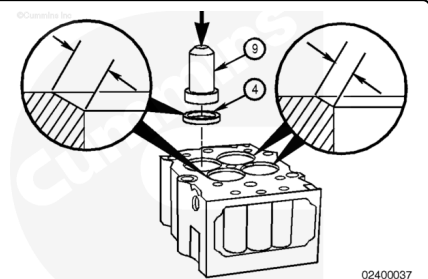


Both the intake and exhaust port valve seat inserts **must** be installed with the 30-degree angle positioned as shown.

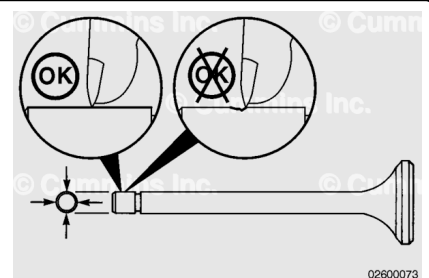
Exhaust valve seat inserts are color coded white to identify the 30-degree angle side of the seat insert.

Note : Exhaust valve seat inserts color-coded Brown have 20-degree angle seats and **must only** be installed to QSK60-G3, G4, and G7 engines.

Use an arbor press and a mandrel to install the valve insert into the head.

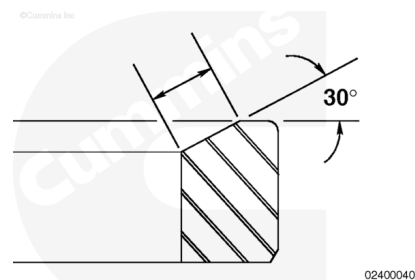


Check the valve tip above the collet groove for damage. Use a fingernail to check for smoothness around all 360 degrees of the valve tip. If any wear can be felt, do **not** reuse the valve. Some appearance of wear that can **not** be felt is normal. The valve tip **must** be smooth. If the valve stem does **not** have a smooth radius, do **not** reuse the valve.



⚠ CAUTION ⚠

Valves seat inserts identified with a yellow paint mark, must not be ground. Grinding these inserts will damage the hardened layer of the insert face and severe engine damage can result.



Use the valve insert seat grinding machine, Part Number ST-685, or equivalent, and the valve guide arbor set, Part Number 3375946, or equivalent, to grind the valve seats.

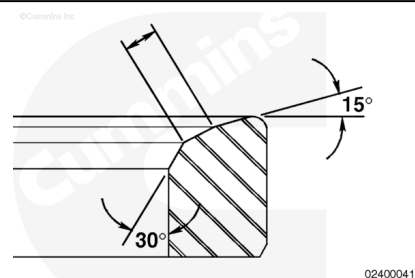
Grind the valve insert seat to the angle shown.

Measure the width.

Valve Insert Seat Width			
	mm		in
Intake	3.05	MIN	0.120
	3.55	MAX	0.140
Exhaust	1.52	MIN	0.060
	2.54	MAX	0.100

If the width of the valve insert seat is **not** within specifications, remove the surface material on the inside diameter and the outside diameter to decrease the width of the seat.

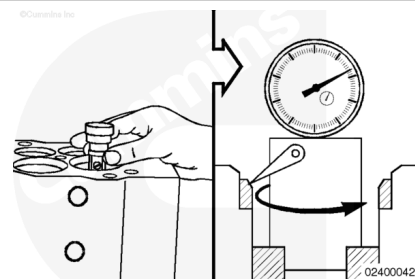
If the valve insert seat specifications are **not** obtained by grinding, the valve insert **must** be replaced.



Measure the valve insert seat to the valve guide concentricity using eccentricimeter, ST-685-4, or equivalent, and the appropriate valve guide arbor, ST-663, or equivalent.

Note : When using the eccentricimeter, the pin that contacts the valve seat **must always** remain on the sealing surface of the seat, or incorrect measurements will be recorded.

The valve insert seat and guide **must** be concentric.



Valve Insert Seat to Guide Concentricity		
mm		in
0.05	MAX	0.002

Use a lead pencil or Prussian blue to mark across the valve face, as shown. Install the valve in the valve guide.

Hold the valve against the valve seat and rotate the valve backward and forward three or four times.

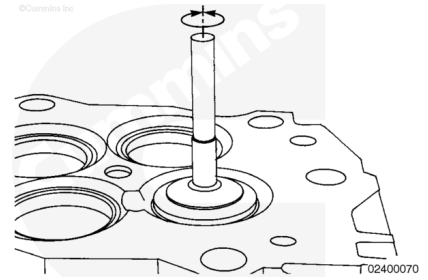
Correct contact against the valve seat will break the marks on the valve face.

Valves and valve seats that are correctly machined do **not** require the use of lapping compound to make an airtight seal. If lapping compound is required, inspect the adjustments of the facing machine and the condition of the grinding stone.

Use a fine lapping compound. Apply a thin and even coating on the valve.

Use a power or a hand suction lapping tool to provide pressure in the center of the valve.

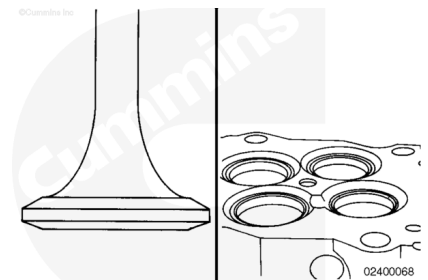
Turn the valve backward and forward. Continue lapping until the compound shows a continuous contact pattern on both the valve seat insert and the valve.



⚠ CAUTION ⚠

Lapping compound is an abrasive material. Failure will result if the cylinder head, valves, and valve seats are not cleaned thoroughly.

Clean the lapping compound from the parts.



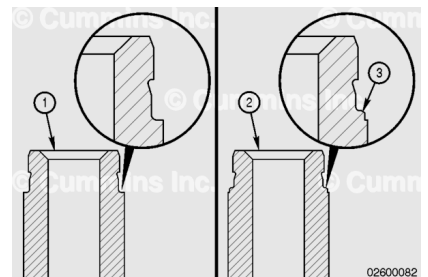
Note : There are two types of valve guides utilized. They are identical apart from the shape at the top of the guide where the valve stem seals are located. It is essential to identify which type of guide is installed and then use the correct procedure to install the valve stem seals.

Valve Guide Type A:

Type A valve stem seals require the use of the depth gauge for installation as detailed in the following steps.

Valve Guide Type B:

Type B valve stem seals can be installed without the use of the depth gauge. Install by pushing the seals onto the guides until they are against the step on the valve guide. The depth



gauge will **not** fit onto the required location of valve guide type B.

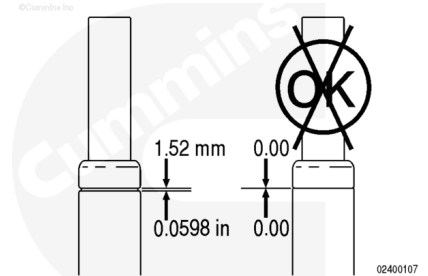
1. Valve guide type A
2. Valve guide type B
3. Step feature. **Only** present on type B valve guides.

⚠ CAUTION ⚠

To avoid the valve seal fitting loosely on the valve guide, do not lubricate the valve seal. The seal must be installed dry.

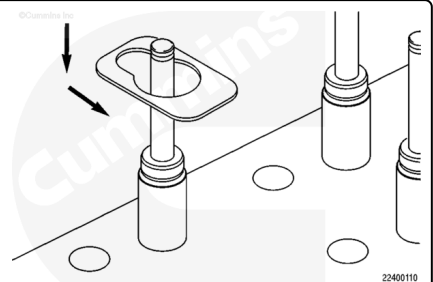
Note : Valve stem seals must be replaced during cylinder head rebuild. Do not use once removed.

If the valve stem seal is installed too far on the guide, the seal will fail prematurely, causing excessive oil consumption. The gauge allows correct positioning of the seal.

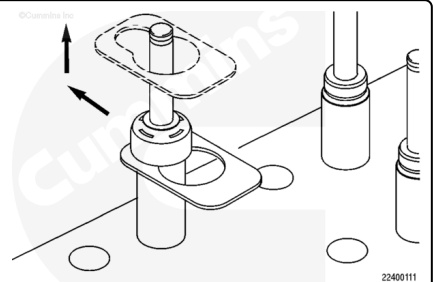


Place the valve stem seal depth gauge over the valve guide and move the smaller slot of the gauge into the valve guide groove area over the shoulder.

Use a 1/2-inch-drive, 5/8-inch or 16-mm socket drive the valve stem seal over the valve guide until it is positioned against the depth gauge, Part Number 3164462.



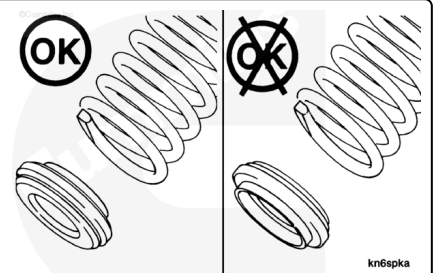
Slide the valve stem seal depth gauge to the large end of the slot and remove the gauge, passing over the valve stem seal outside diameter.



When turning the rotator by hand, it can turn roughly, feel difficult to turn, or **not** turn at all. This is normal until the rotator is installed; after installation it will rotate freely.

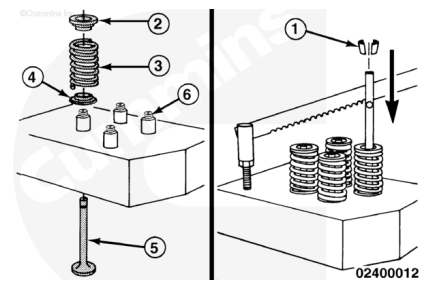
The rotator **must** be soaked in clean engine oil for at least 15 minutes prior to installation.

The rotator **must** be installed over the valve guide with the spring pilot flange face-up, as shown in this illustration.



⚠ WARNING ⚠

The valve springs are under compression. Use caution when using the valve spring compressor. Personal injury can result.



Use valve spring compressor, Part Number 3163606, 3375960, or equivalent, for single valve installation. The valve spring stand, Part Number ST-1022, and compressor plate, Part Number 3133177, can be used to compress all four springs at the same time.

The valve rotator **must** be soaked in clean engine oil for at least 15 minutes prior to installation.

Use clean engine oil to lubricate the valve stems prior to installation.

Assemble the parts in the following order:

- Valve (5)
- Valve rotator (4)
- Valve stem seal (6)
- Valve spring (3)
- Valve spring retainer (2)
- Two new valve collets (1).

Compress the valve spring and install the following parts:

- New valve collets (1).

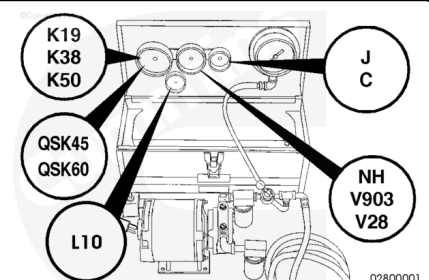
Vacuum Test

The valve vacuum tester, Part Number 3824277 (115-VAC, 50/60 Hz) or 3824278 (220-VAC, 50/60 Hz), can be used to test all Cummins® engine models.

Before using the tester, test the leakage shutoff valve. If the valve is dirty or worn, it will produce a false leakage measurement on the gauge.

Check the valve:

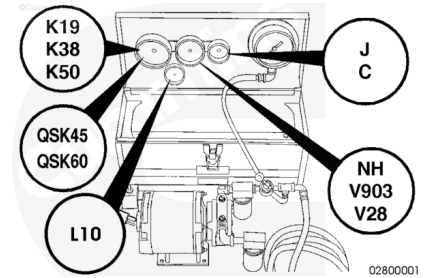
- Open the shutoff valve.
- Turn on the vacuum pump.
- Place the cup against a smooth surface.
- Close the shutoff valve.



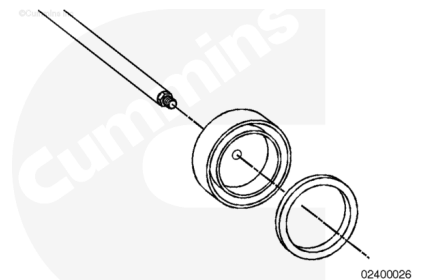
- Turn off the vacuum pump.
- Wait approximately 10 seconds.
- The gauge **must not** drop more than 7 kPa [2 in Hg].

The valve and the valve seats **must** be clean and dry.

Choose the correct cup and the correct seal from the service tool kit for the engine model that is to be tested.

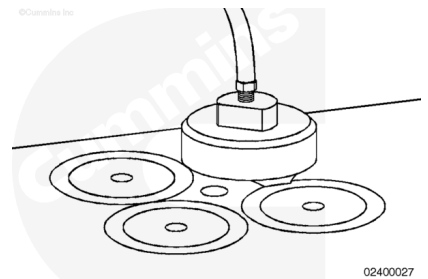


Install the seal and the cup to the vacuum line (hose).



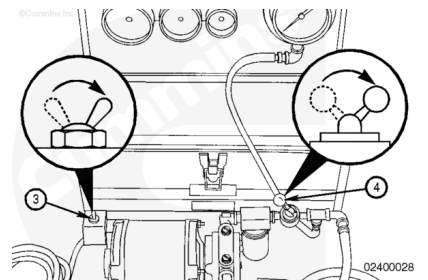
Cover the valve with the cup and the seal. The seal **must** have a tight contact on the cylinder head around the valve.

To check the exhaust valves, the seal **must** completely fill the milled area between the exhaust valves.



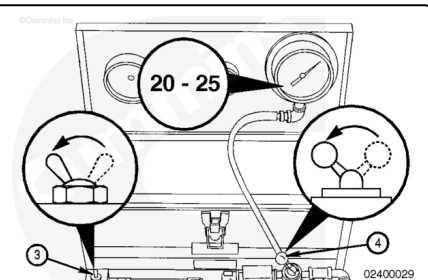
Move the toggle switch (3) to the ON position.

Turn the vacuum control valve (4) to the OPEN position.



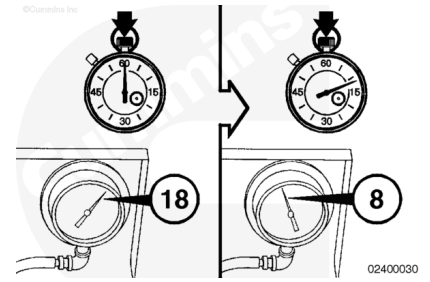
When the gauge indicates between 34 to 85 kPa [10 to 25 in Hg], turn the vacuum control valve (4) to the CLOSED or OFF position.

Turn the toggle switch (3) to the OFF position.



⚠ CAUTION ⚠

The cylinder head must be disassembled and cleaned after any grinding or cutting procedures to reduce the possibility of engine damage.



Use a stopwatch. As the vacuum gauge needle moves **counterclockwise**, start timing when the needle on the gauge points to 61 kPa [18 in Hg].

Stop timing when the needle on the gauge points to 27 kPa [8 in Hg].

The elapsed time for the gauge to move between the specified readings **must** be 10 seconds or more.

If the elapsed time is less than 10 seconds, perform the following checks:

- Repeat the test to be certain the equipment is functioning properly.
- Use a mallet to lightly hit the valve stem to be certain the valve is sealed. Repeat the test.
- Apply a thin coating of grease on the outside diameter of the insert and the valve head. Repeat the vacuum test. The grease pattern will show the point of leakage.
- If the leakage is between the valve insert and the head, the insert **must** be replaced.

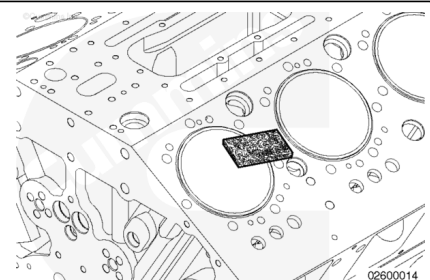
If the leakage is between the valve and the valve insert seat, one of the following procedures **must** be performed:

- Lap valve to insert seat
- Grind the valves
- Grind the valve insert seat.

Install

⚠ CAUTION ⚠

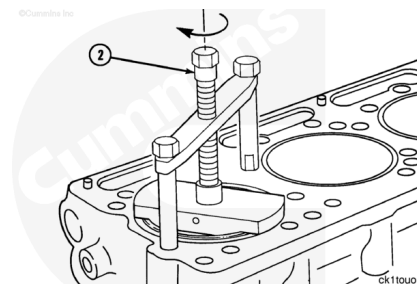
Using power or pneumatic tools to clean the cylinder head gasket surface on the cylinder block can damage the cylinder liners.



Clean the top of the cylinder block and the cylinder liners.
Refer to Procedure 001-026 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-026.html)

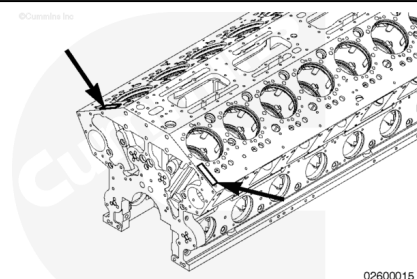
Measure the cylinder liner protrusion. Refer to Procedure 001-064 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-064.html)

If the cylinder liner protrusion is **not** correct, it **must** be adjusted before installing the cylinder head. Refer to Procedure 001-028 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-028-tr.html)



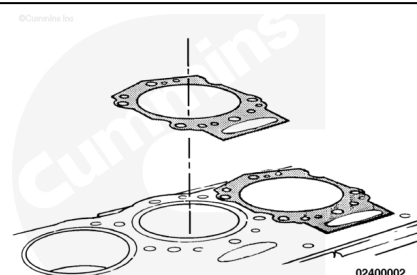
Inspect the cylinder block to determine if an oversize or a standard size head gasket is required.

An oversize head gasket is often required on blocks that have had material removed from the top surface of the block. Check the block for markings indicating an oversize gasket is required.



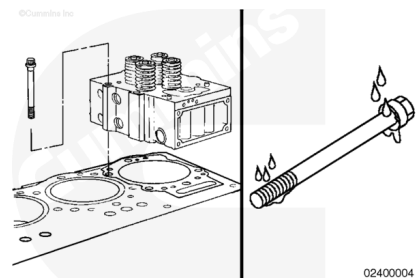
The word TOP, stamped on top of the head gasket, **must** be visible after the gasket is installed.

Install the cylinder head gasket, use the dowel pins for alignment.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



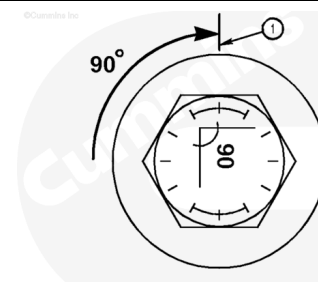
Install the cylinder head. It **must** slide easily over the dowel pins.

Lubricate the capscrew head flange with clean engine oil.

Lubricate the capscrew threads with clean engine oil. Allow excess oil to drip off the capscrews before installing them in the block.

Install the capscrews.

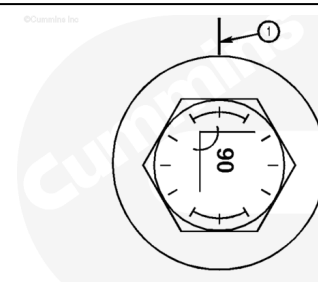
When using torque plus angle, the tolerance on the 90-degree angle of rotation is 1½ flats (90 degrees ±5 degrees). If the capscrew is rotated beyond two flats, do **not** loosen the capscrew. The clamp load is still acceptable; however, rotating the capscrew beyond two flats causes additional stretch and reduces the life of the capscrew. With proper tightening, the capscrew can typically be reused for the life of the engine.



02600016

After all capscrews on one head have been tightened to 300 N•m [221 ft-lb], they **must** be rotated in the tightening direction an additional 90 degrees. Rotate the capscrew until the mark on the cylinder head (1) is between the next two marks joined by an arc (more than one flat and less than two flats).

The markings on the head of the flange head capscrews serve as an aid during installation. After tightening the capscrew to 300 N•m [221 ft-lb], mark the cylinder head (1) adjacent to one of the two single marks on the capscrew head.

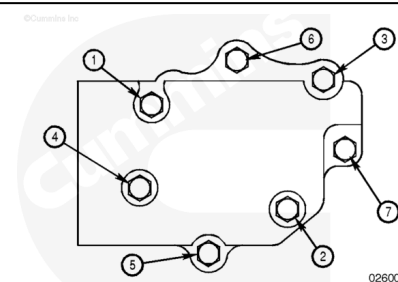


02600017

Tighten the cylinder head mounting capscrews, use the following steps and in the sequence shown.

Torque Value:

1. 70 N•m [52 ft-lb]
2. 200 N•m [148 ft-lb]
3. 300 N•m [221 ft-lb]
4. Turn another 90 degrees (clockwise)



02600018

Finishing Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

- Install the injectors. Refer to Procedure 006-026 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)
- Install the rocker lever housings. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-013-tr.html)
- Install the push rods. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html)
- Install the crossheads. Refer to Procedure 002-001 in Section 2.
(/qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html)
- Install the rocker levers. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html)
- Adjust the valves and injectors. Refer to Procedure 003-006 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-006-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html)
- Install the fuel manifold. Refer to Procedure 006-022 (Fuel Manifold in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-022-tr.html)
- Install the fuel drain lines. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-013-tr.html)
- Install the aftercooler assembly and the air crossover. Refer to Procedure 010-002 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html)
- Connect the wiring harness. Use this procedure from the Troubleshooting and Repair Manual, QSK38, QSK50, and QSK60, Bulletin 4021533. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-043.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/56/56-013-009.html)
- Connect the air supply to the air starter, if installed.
- Fill the cooling system(s). Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Operate the engine to 71°C [160°F] minimum coolant temperature and check for leaks.

QSK60 Marine Applications

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

Do not use gasket cement. It will prevent the gasket from sealing properly. Edge-molded gaskets do not require sealant.

- Install the injectors. Refer to Procedure 006-026 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)
- Install the rocker lever housings. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-013-tr.html)
- Install the push rods. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html)
- Install the crossheads. Refer to Procedure 002-001 in Section 2.
(/qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html)
- Install the rocker levers. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html)
- Adjust the valves and injectors. Refer to Procedure 003-006 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-006-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html)
- Install the fuel manifold. Refer to Procedure 006-022 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-022-tr.html)
- Install the aftercooler assembly and the air crossover. Refer to Procedure 010-002 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html)
- Connect the wiring harness. Use this procedure from the Troubleshooting and Repair Manual, QSK38, QSK50, and QSK60, Bulletin 4021533. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-043.html)
- Install the exhaust manifold heat shields. Refer to Procedure 011-015 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-015.html)
- Install the turbocharger heat shields. Refer to Procedure 010-076 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-076-tr.html)
- Install the aftercooler heat shields. Refer to Procedure 010-129 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/56/56-013-009.html)
- Connect the air supply to the air starter, if installed.

- Fill the cooling system(s). Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Operate the engine to 71°C [160°F] minimum coolant temperature and check for leaks.

with Electronically Actuated Injector

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first, and attach the negative (-) battery cable last.

- Install the rocker lever housings. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-013-tr.html)
- Install the injectors. Refer to Procedure 006-026 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)
- Install the push rods. Refer to Procedure 004-014 in Section 4.
(/qs3/pubsys2/xml/en/procedures/56/56-004-014-tr.html)
- Install the crossheads. Refer to Procedure 002-001 in Section 2.
(/qs3/pubsys2/xml/en/procedures/56/56-002-001-tr.html)
- Install the rocker levers. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html)
- Adjust the valves. Refer to Procedure 003-006 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-006-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html)
- Install the injector supply lines. Refer to Procedure 006-051 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-051-tr.html)
- Install the fuel drain lines. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/56/56-006-013-tr.html)
- Install the aftercooler assembly and the air crossover. Refer to Procedure 010-002 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/56/56-011-007-tr.html)
- Connect the wiring harness. Use this procedure from the Troubleshooting and Repair Manual, QSK38, QSK50 and QSK60 (CM850 Modular Common Rail System), Bulletin 4021533. Refer to Procedure 019-043 in Section 19.
(/qs3/pubsys2/xml/en/procedures/05/05-019-043.html)

- Connect the batteries. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/56/56-013-009.html)
- Connect the air supply to the air starter, if installed.
- Fill the cooling system(s). Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Operate the engine to 71°C [160°F] minimum coolant temperature. Check for leaks.

Last Modified: 20-Mar-2026
